

Online Appendix for “Pairs Trading with Wavelet Transform”

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1 Johansen’s Cointegration test

The Johansen’s (1988) Cointegration test can be used to reveal the presence of long run relation among a set of variables. Let $\{y_t\}_{t=1}^T$ be a p column vector of the observed time series, which are $I(1)$. In our case, $y_t = [S_{i,t}, S_{j,t}]^T$ for some pairs of price series and $p = 2$. We say that the variables in y_t are cointegrated with rank r , if there exist a linearly independent full rank $p \times r$ matrix γ such that $\gamma^T y_t$ is $I(0)$. If $r = 0$, the variables in y_t does not constitute a cointegrating relation, whereas $r > 0$ indicates at least one cointegrating relation among these series. For instance, $r = 1$ indicates that there exists one linear relation which establishes the long run comovement among the variables in y_t .

In general, we would like to test the existence of a specific number of cointegrating relations versus the alternative of a higher number of relations. We can demonstrate these hypothesis as $H_0 : r = r_0$ versus $H_a : r > r_0$ for $r_0 = 0, 1, \dots, p - 1$. We follow a sequential procedure for the cointegration rank selection in the testing framework. First, we test the absence of cointegration, $H_0 : r = 0$. If we cannot reject the null, we say that the prices of the securities are not cointegrated. Otherwise, we test $H_0 : r = 1$, and if we do not reject the null, we say that the prices of the securities are cointegrated of rank one. As a result, we can use these pairs in the trading exercise. Note that there can be at most one cointegrating relation with two price series, thus, we stop the testing sequence at $r = 1$.

2 DWT & MODWT: Technical Explanations

We define the DWT first, then show how we obtain the MODWT from the specifications of the DWT. We note that what follows is outlined from Gençay et al. (2001), Fan and Gençay (2006), and Conlon et al. (2018).

The essence of the DWT consists of a pair of filters, namely high- and low-pass filters. The high-pass filter is for modeling the short-run fluctuations, whereas the low-pass filter is for the long-run fluctuations. The filters are the oscillating coefficients and for the formal mathematical formulation, let $h = (h_0, h_1, \dots, h_{L-1})$ be the high-pass filter with filter length L . Then, we write the following properties:

$$(i) \sum_{l=0}^{L-1} h_l = 0, \quad (ii) \sum_{l=0}^{L-1} h_l^2 = 1, \quad \text{and} \quad (iii) \sum_{l=-\infty}^{\infty} h_l h_{l+2n} = 0 \text{ for all integers } n \neq 0.$$

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⁴Here the sign $\top$ denotes the transpose.

In the above, the filter length L determines the lag length of the filtration. The first and second properties indicate that the elements of the high-pass filter sum to zero with unit energy. More clearly, as explained in Fan and Gençay (2006), property (ii) suggests excursions around zero, while property (i) suggests such excursions must eventually cancel out; hence the wave property of the short-run frequency of the decomposition. The third property implies the orthogonality of the high-pass filter to even shifts. Along with the second property, the third property assures that the length of the high-pass filter is equal to that of the low-pass filter $g = (g_0, g_1, \dots, g_{L-1})$ which is the local averaging operator. The coefficient values of the low-pass filter are found through the quadrature mirror relation, $g_l = (-1)^{l+1} h_{L-1-l}$ for $l = 0, 1, \dots, L-1$. Then, it follows that the low-pass filter satisfies the properties:

$$(i) \sum_{l=0}^{L-1} g_l = \sqrt{2}, \quad (ii) \sum_{l=0}^{L-1} g_l^2 = 1 \quad \text{and} \quad (iii) \sum_{l=-\infty}^{\infty} g_l g_{l+2n} = 0 \text{ for all integers } n \neq 0,$$

and

$$(iv) \sum_{l=0}^{L-1} g_l h_{l+2n} = \sum_{l=-\infty}^{\infty} g_l h_{l+2n} = 0 \text{ for all integers } n \neq 0.$$

The property (iv) is also known as the orthogonality to even shifts of complementary filters because the low-pass filter complements the high-pass filter. Clearly, this property follows from the quadratic mirror relation and establishes a certain level of independence between two complementary filters.

Next, to obtain the filtered series under the MODWT, we write the following:

$$\tilde{h}_l = \frac{1}{\sqrt{2}} h_l \quad \text{and} \quad \tilde{g}_l = \frac{1}{\sqrt{2}} g_l,$$

where $\tilde{h}, \tilde{g}$ are the high and low-pass filters respectively under the MODWT method. We utilize these filters with the convolution operator and the Pyramid algorithm of Mallat (1989) to obtain MODWT coefficients. To this end, we consider a time series of interest $(Z_t)_{t \in \{1, \dots, T\}}$ with dyadic length $T = 2^U$. We can apply the high and low pass filters to the observed series in order to obtain the MODWT coefficients for *Level 1* by

$$\tilde{V}_{1,t} = \sum_{l=0}^{L-1} \tilde{g}_l Z_{t-l \bmod T}, \quad \tilde{W}_{1,t} = \sum_{l=0}^{L-1} \tilde{h}_l Z_{t-l \bmod T}. \quad (1)$$

In the above equation, the MODWT coefficients $\tilde{V}_{1,t}$ and $\tilde{W}_{1,t}$ are obtained by convolving the low- and high-pass filters with the times series $(Z_t)_{t \in \{1, \dots, T\}}$, respectively; hence the summations we see in Equation (1). The Pyramid Algorithm, on the other hand, implies further convolution of $\tilde{V}_{k,t}$ component for $k = 1, \dots, J^*$ with the low- and high-pass filters. More clearly, once we obtain $\tilde{V}_{1,t}$ under Level 1 as in (1), then we do the same operations to $\tilde{V}_{1,t}$ instead of convolving with $(Z_t)_{t \in \{1, \dots, T\}}$ under Level 2, and the same continues up until the level of transformation J^* . Given these, then, the formal mathematical exposition follows

$$\tilde{V}_{k,t} = \sum_{l=0}^{L-1} \tilde{g}_l \tilde{V}_{k-1,t-l \bmod T}, \quad \text{and} \quad \tilde{W}_{k,t} = \sum_{l=0}^{L-1} \tilde{h}_l \tilde{V}_{k-1,t-l \bmod T} \quad \text{for } k = 2, \dots, J^*. \quad (2)$$

Notice that each of the series $(\tilde{V}_{k,t})_{t \in \{1, \dots, T\}}$ and $(\tilde{W}_{k,t})_{t \in \{1, \dots, T\}}$ generated in Equations (1) and (2) has the same length as the observable variable $(\tilde{Z}_t)_{t \in \{1, \dots, T\}}$.

3 Mean Profits from Hodrick-Prescott Filters

Table 1: Mean Returns: Hodrick-Prescott Filters

	Training	2010-2011	2011-2012	2012-2013	2013-2014	2014-2015	2015-2016	2016-2017
	Trading	2011-2012	2012-2013	2014-2015	2014-2015	2015-2016	2016-2017	2017-2018
Cointegration	m-hp	-1.53%	-4.35%	-7.42%	0.10%	0.54%	-4.41%	-2.75%
	hp(10)	-6.51%	-7.43%	-10.34%	-4.66%	-2.24%	-6.77%	-4.94%
	hp(50)	-8.29%	-8.39%	-11.34%	-7.17%	-3.06%	-7.81%	-5.72%
	hp(100)	-9.05%	-8.84%	-11.96%	-8.05%	-3.37%	-8.27%	-6.00%
Minimum Distance	m-hp	0.14%	-2.71%	-2.18%	-1.21%	-0.29%	-0.57%	-4.31%
	hp(10)	-4.39%	-5.94%	-5.27%	-4.30%	-4.27%	-3.79%	-6.86%
	hp(50)	-5.82%	-7.14%	-6.46%	-5.22%	-5.69%	-5.20%	-7.79%
	hp(100)	-6.51%	-7.70%	-6.95%	-5.69%	-6.26%	-5.88%	-8.18%

Note: **m-hp** stands for the modified Hodrick-Prescott filter of ?. **hp(λ)** stands for the standard Hodrick-Prescott filter with smoothing parameter λ .

4 Yearly Key Measures Before Transaction Costs

In Tables 2 and 3, we report the yearly evolution of summary statistics regarding the returns from the methods employed in the study. These statistics correspond to the total profits at the end of each trading period. For example, the mean is the sample mean of the total profits of the number of pairs considered (active and non-active) during a trading period. We first report the results from the standard spread in Table 2, then from the *sym22* wavelet filtered spread in Table 3.

Table 2: Summary Statistics for Excess Returns from Standard Spread

Method	Spread	Training	2010-2011	2011-2012	2012-2013	2013-2014	2014-2015	2015-2016	2016-2017
		Trading	2011-2012	2012-2013	2014-2015	2014-2015	2015-2016	2016-2017	2017-2018
Cointegration	Standard	Mean	-0.0431%	-3.4232%	-6.4979%	1.4279%	1.6106%	-3.6206%	-2.1172%
		Mean ($t + 1$)	-1.37%	-3.45%	-6.77%	1.03%	1.49%	-3.46%	-1.93%
		t-stat	-0.0262	-6.2015	-7.6814	1.2447	2.8153	-4.6155	-3.0927
		Std. Dev.	1.1317	0.4003	0.4412	0.6896	0.4316	0.5444	0.4107
		Skewness	-48.5924	-2.3112	-1.3574	26.4252	-2.2355	-7.2575	4.3541
		Kurtosis	2963.1126	59.4275	48.6905	1289.1850	68.5051	261.0167	118.2725
		p-value(JB)	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010
Minimum Distance	Standard	Mean	1.2398%	-1.7999%	-1.0669%	-0.3254%	0.7991%	0.6939%	-3.4212%
		Mean ($t + 1$)	0.85%	-1.69%	-0.93%	-0.63%	0.25%	0.66%	-3.23%
		t-stat	1.0575	-1.4162	-1.1006	-0.3984	0.9098	0.4806	-4.4123
		Std. Dev.	0.3707	0.4019	0.3065	0.2583	0.2778	0.4565	0.2452
		Skewness	-4.4748	-7.5853	3.7158	3.5811	1.1456	-2.7014	0.3561
		Kurtosis	85.3942	132.7057	46.1025	54.8995	11.2337	53.5658	12.8574
		p-value(JB)	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010

Table 3: Summary Statistics for Excess Returns from *sym22* Spread

Method	Spread	Training	2010-2011	2011-2012	2012-2013	2013-2014	2014-2015	2015-2016	2016-2017
		Trading	2011-2012	2012-2013	2014-2015	2014-2015	2015-2016	2016-2017	2017-2018
Cointegration	Sym22	Mean	13.5274%	10.0859%	10.6988%	4.7980%	8.2136%	8.7794%	11.5032%
		Mean ($t + 1$)	13.20%	10.17%	10.46%	4.90%	8.22%	8.61%	11.29%
		t-stat	22.0777	23.9500	11.9871	7.1783	15.9023	15.0542	18.4619
		Std. Dev.	0.4218	0.3054	0.4655	0.4018	0.3897	0.4048	0.3738
		Skewness	15.0515	3.8834	-8.3035	-3.4344	-4.1232	-3.0045	2.3835
		Kurtosis	494.6974	56.3641	273.9854	49.8503	71.7888	242.9690	34.7098
		p-value(JB)	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010
Minimum Distance	Sym22	Mean	13.58%	10.93%	14.69%	7.97%	13.63%	10.62%	11.29%
		Mean ($t + 1$)	13.77%	10.75%	14.60%	8.10%	13.67%	10.50%	11.17%
		t-stat	1.0575	-1.4162	-1.1006	-0.3984	0.9098	0.4806	-4.4123
		Std. Dev.	0.3707	0.4019	0.3065	0.2583	0.2778	0.4565	0.2452
		Skewness	-4.4748	-7.5853	3.7158	3.5811	1.1456	-2.7014	0.3561
		Kurtosis	85.3942	132.7057	46.1025	54.8995	11.2337	53.5658	12.8574
		p-value(JB)	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010	0.0010

In Tables 4 and 5, we report the yearly evolution of results regarding the downside risk measures. We report the values from the standard spread approach in Table 4, and those from the *sym22* spread approach in Table 5.

Table 4: Downside Risk Measures of The Trading Period Profits from Standard Spread

Method	Spread	Training	2010-2011	2011-2012	2012-2013	2013-2014	2014-2015	2015-2016	2016-2017
		Trading	2011-2012	2012-2013	2014-2015	2014-2015	2015-2016	2016-2017	2017-2018
Cointegration	Standard	% positive	56.19%	41.86%	42.87%	53.68%	56.04%	44.82%	47.67%
		VaR (5%)	-58.15%	-50.57%	-62.49%	-51.91%	-43.83%	-41.47%	-50.26%
		CVaR (5%)	-153.99%	-101.45%	-111.61%	-95.75%	-96.45%	-107.41%	-87.01%
		Mean-to-CVaR	0.0003	0.0337	0.0582	-0.0149	-0.0167	0.0337	0.0243
Minimum Distance	Standard	% positive	58.60%	50.00%	47.50%	52.10%	51.40%	56.50%	47.70%
		VaR (5%)	-42.33%	-44.99%	-41.54%	-34.04%	-37.28%	-38.35%	-43.80%
		CVaR (5%)	-90.32%	-99.09%	-60.26%	-55.71%	-58.26%	-109.13%	-59.95%
		Mean-to-CVaR	-0.0137	0.0181	0.0177	0.0058	-0.0137	-0.0064	0.0571

Note: In the table % positive denotes the percentage of positive returns.

Table 5: Downside Risk Measures of The Trading Period Profits from Sym22 Spread

Method	Spread	Training	2010-2011	2011-2012	2012-2013	2013-2014	2014-2015	2015-2016	2016-2017
		Trading	2011-2012	2012-2013	2014-2015	2014-2015	2015-2016	2016-2017	2017-2018
Cointegration	Sym22	% positive	71.51%	57.47%	67.46%	65.63%	71.08%	59.71%	69.11%
		VaR (5%)	-18.34%	-17.09%	-22.53%	-41.84%	-37.68%	-16.60%	-24.59%
		CVaR (5%)	-45.47%	-40.12%	-71.41%	-109.49%	-91.65%	-55.70%	-59.44%
		Mean-to-CVaR	-0.2975	-0.2514	-0.1498	-0.0438	-0.0896	-0.1576	-0.1935
Minimum Distance	Sym22	% positive	74.90%	71.90%	79.90%	77.10%	74.40%	74.80%	78.00%
		VaR (5%)	-10.77%	-13.62%	-10.69%	-11.26%	-12.39%	-14.53%	-10.74%
		CVaR (5%)	-23.59%	-25.88%	-23.92%	-39.87%	-24.10%	-61.70%	-30.52%
		Mean-to-CVaR	-0.5756	-0.4222	-0.6139	-0.2000	-0.5656	-0.1721	-0.3700

Note: In the table % positive denotes the percentage of positive returns.

In Table 6, we provide further statistics regarding the downside risk measures. The values in the previous tables are obtained from the trading period end profits of the pairs, while the measures in the table below are obtained from the trading period time series evolution of returns. The first one is the celebrated Sharpe ratio, which is a symmetric measure of returns' deviations from their mean values. For the second measure, we consider the maximum drawdown. While the Sharpe ratio is annualized, maximum drawdown is given daily, because it provides the maximum potential loss in a given day during the course of a trading period. The value is reported with a positive sign.

Table 6: Downside Risk Measure from The Time Series Evolution of Returns

Method	Spread	Training	2010-2011	2011-2012	2012-2013	2013-2014	2014-2015	2015-2016	2016-2017
		Trading	2011-2012	2012-2013	2014-2015	2014-2015	2015-2016	2016-2017	2017-2018
Cointegration	Standard	Sharpe Ratio	-0.0051	-0.8594	-1.2644	0.2270	0.5659	-0.9558	-0.5390
		Max-drawdown	5.78%	5.36%	7.43%	5.97%	2.06%	5.60%	6.91%
	Sym22	Sharpe	1.4652	5.4395	3.3760	1.7078	2.9240	2.5598	2.2669
		Max-drawdown	4.75%	0.65%	1.65%	2.52%	0.93%	2.60%	2.98%
Minimum Distance	Standard	Sharpe Ratio	0.3530	-0.6018	-0.3248	-0.1109	0.1826	0.2005	-1.1908
		Max-drawdown	4.39%	2.50%	3.91%	4.87%	3.14%	3.44%	5.43%
	Sym22	Sharpe	2.4752	5.2853	4.3223	2.8603	3.1500	3.8819	3.8726
		Max-drawdown	2.49%	0.82%	1.30%	1.15%	1.65%	1.26%	1.65%

5 Simulation Details

In the simulation analysis, we start by constructing two latent processes that are related to each other through a spread type specification considered in the study. We call this relation “the true long-run relation” in between these processes. We later contaminate these series with error processes which are similar to measurement errors, and see how OLS regression of the contaminated series work before and after wavelet application.

First, we consider a pair of latent price processes $\{(S_{1,t}^*, S_{2,t}^*)\}_{t=1}^T$ for two securities. These latent prices are used to denote the long-run relation between these securities. Notice that we do not

assume, a priori, cointegration between these variables, but they are connected with a linear relation of the form

$$S_{1,t}^* = \alpha^* + \beta^* S_{2,t}^* + u_t^*; \quad (3)$$

$$S_{2,t}^* = \Delta^{-d_2} u_{2,t}^*, \quad (4)$$

where Δ^{-d_2} , with $d_2 \in [1/2, 1]$ (we set $d_2 = 1$ in the simulation), is the fractional integration operator, $u_{2,t}^* \sim N(0, \sigma_2^2)$ and $u_t^* \sim I(d)$ with $d \in [0, d_2]$ (i.e. $(u_t^*)_{t=1}^T$ is a fractionally integrated process). Here as well, we set $d = 0$ and $d = 1$ later; the first choice reflects cointegration, while the second one reflects the case of related but not cointegrating variables. With our choices for d_2 and d , we aim to construct a simulation scenario in line with what we analysed in the main sequel. Nonetheless to say, with other choices, the results we obtain here do not change.

In the simulations, the true parameter values are given as α^* and β^* . In the simulation, we assign values exogenously to these parameters, then contaminate the pair $\{(S_{1,t}^*, S_{2,t}^*)\}_{t=1}^T$, and see how well the regular and wavelet based OLS regressions perform in approximating β^* , which is the parameter of interest in our study. Thus, as stated above, our aim is to understand how well we capture the true relation between two non-stationary price series after the application of wavelets.

To this end, we contaminate the above latent price series at the high frequency (short-run/noise) level. In other words, we create an observed price sequence pair $\{(S_{1,t}, S_{2,t})\}_{t=0}^T$ through the procedure

$$S_{1,t} = \sum_{l=0}^{L-1} g_l^* S_{1,t}^* + \sum_{l=0}^{L-1} h_l^* \epsilon_{1,t}^*; \quad (5)$$

$$S_{2,t} = \sum_{l=0}^{L-1} g_l^* S_{2,t}^* + \sum_{l=0}^{L-1} h_l^* \epsilon_{2,t}^*, \quad (6)$$

where $\{(g_l^*, h_l^*)\}_{l=0}^{L-1}$ is a pair of *inverse* low and high-pass filters that effectively combine the different frequency components. In this way, we construct the observed price series. In this construction, $[\epsilon_{1,t}^*, \epsilon_{2,t}^*]^\top \sim D(0_{2 \times 1}, \text{diag}\{\sigma_1^2, \sigma_2^2\}^\top)$ is vector of the contaminating errors. From the construction, we see that the procedure followed here is the backward of what has been followed in the main study. There, we derive the long-run components from the observed and supposedly contaminated price series, then run a wavelet based regression model to estimate the parameter β_w that is shown to be the main source of improvement in our results. Here, we construct the price series by assigning values to parameters exogenously, then contaminate them with the expectation that wavelet based regression technique will provide better performance in approximating the parameter of interest.

To achieve our goal, we analyse the results from two regression models:

$$S_{1,t} = \alpha + \beta S_{2,t} + u_t; \quad (7)$$

$$V_{1,t} = \alpha_w + \beta_w V_{2,t} + e_t, \quad (8)$$

where $V_{1,t}$ and $V_{2,t}$ are the long-run coefficients from the MODWT of the contaminated series $S_{1,t}$ and $S_{2,t}$, respectively. Suppose that $\hat{\alpha}$ and $\hat{\beta}$ are the OLS estimators from Equation 7, and $\hat{\alpha}_w$ and $\hat{\beta}_w$ are the OLS estimators from Equation 8. That is, the first model is for the results from OLS estimation under the standard approach, and the latter is from the wavelet based OLS estimation. In our simulations, we use these models to check for two claims:

- The wavelet-based slope estimator $\hat{\beta}_w$ is less biased than the standard slope estimator $\hat{\beta}$: $|\beta^* - E(\hat{\beta}_w)| < |\beta^* - E(\hat{\beta})|$.

- The wavelet-based slope estimator $\hat{\beta}_w$ generates less mean squared forecast error than the standard slope estimator $\hat{\beta}$: $E \left[(S_1 - \hat{\alpha}_w - \hat{\beta}_w S_2)^2 \right] < E \left[(S_1 - \hat{\alpha} - \hat{\beta} S_2)^2 \right]$.

The difficulty of analysis with wavelets is the lack of their clear analytical tractability. For this reason, a response to above claims through analytical formulation is not very straightforward and requires a standalone study; we skip the formal proof by leaving the task to a future study and provide a brief simulation to give some insight about the problem. We believe that parallel results between here and the paper would indeed provide an insight.

In our simulation exercise, we set the sample size $T = 250$ for connecting the simulation results with our empirical findings. We generate the data according to Equations (3)-(6) with different parameter values $\theta = (d, \beta^*, \sigma_1^2, \sigma_2^2)$ and $2T$ observations. Additionally, we simulate $S_{2,t}$ as a random walk process, hence $d_2 = 1$. Likewise $S_{1,t} \sim I(1)$. Then, as mentioned above, the error of the Regression Equation 3, u_t^* is either cointegrating ($u_t^* \sim I(0)$) error term, or related but not cointegrating ($u_t^* \sim I(1)$) error term. The left panel of Table 7 contains the results from the first case ($d = 0$), and the right panel contains the second one ($d = 1$).

In Table 7, we depict some crucial statistics obtained from these simulations. In these simulations, notice from the same table that, we consider $\beta^* = 1, 2, 4$ as true values for the relationship in Equation 3. We may also observe that the values assigned for σ_1^2 and σ_2^2 are 1.581, 7.908, 15.811 and 31.263. When reporting, we select values for σ_2^2 one-by-one, then place the values of σ_1^2 next to a selected value of σ_2^2 . In this way, we are able to observe how the results change with changing contamination levels.

When reporting the results, we first consider the average OLS estimates obtained from the standard prices, and, then, their wavelet filtered versions in the training period $t = 1, \dots, T$. These, respectively, correspond to values given under columns $\hat{\beta}$ and $\hat{\beta}_w$. For wavelet filtering, we use *sym22* filter, but other filters such as *db6* also generates similar results. We omit the results for the intercept terms, as the parameter of interest in the main sequel is the slope term. Next, we compute the mean squared forecast errors for both methods. We utilize the following equation to obtain these values:

$$MSFE_s = \sum_{t=T+1}^{2T} \left(S_{1,t} - \hat{\alpha} - \hat{\beta} S_{2,t} \right)^2 \quad \text{and} \quad MSFE_w = \sum_{t=T+1}^{2T} \left(S_{1,t} - \hat{\alpha}_w - \hat{\beta}_w S_{2,t} \right)^2$$

where the subscript s stands for the standard model.

From the results in Table 7, we observe few important features. First, the wavelet-based estimator $\hat{\beta}_w$ suffers minimal amount of bias while the standard estimator $\hat{\beta}$ deviates more from the true value β^* . This deviation is almost monotonic with the increase in the variance of the second contamination noise (i.e. σ_2^2). Second, in every case, the wavelet-based method generates less MSFE. This finding implies that within a trading period, the wavelet-based method generates less divergent behavior than the standard method. The gap between two methods is more nuanced with higher values of the variance of the second contamination noise. Third, with higher values of β^* , we observe higher bias in the standard method. This outcome is also apparent in the MSFE. However, there is an additional pattern for this statistic. The error-noise variance of the standard price series seem to have a higher influence on the MSFE.

In sum, our findings from the simulation study also show the frequency domain measurement error-bias. Particularly, from the results, we see that when wavelets clear out the high frequency (problematic) components, they lead to more desirable estimates for the slope parameter. In this context, the true slope coefficient β^* governs a long run (or low frequency) relation. The deviation from this parameter value can also lead a divergent behaviour for the spread process. Hence, correcting the bias in this parameter leads better trading performance.

Table 7: Simulation Results with High Frequency Contamination

			$S_1 = \alpha^* + \beta^* S_2 + \epsilon \quad \epsilon \sim I(0)$				$S_1 = \alpha^* + \beta^* S_2 + \epsilon \quad \epsilon \sim I(1)$			
β^*	σ_2^2	σ_1^2	$\hat{\beta}$	$\hat{\beta}_w$	$MSFE_s$	$MSFE_w$	$\hat{\beta}$	$\hat{\beta}_w$	$MSFE_s$	$MSFE_w$
1	1.581	1.581	0.968	0.998	1.452	0.749	0.965	0.993	7.705	6.946
		7.906	0.967	0.998	2.296	0.856	0.983	1.013	7.903	6.929
		15.811	0.967	0.998	3.039	0.972	0.985	1.019	8.248	6.982
		31.623	0.967	0.999	4.140	1.172	0.962	0.991	8.752	7.028
	7.906	1.581	0.862	0.991	2.328	0.858	0.856	0.981	7.874	6.947
		7.906	0.864	0.991	2.942	0.952	0.873	1.001	8.125	7.021
		15.811	0.865	0.991	3.549	1.057	0.863	0.989	8.255	6.896
		31.623	0.863	0.991	4.531	1.245	0.873	1.006	8.781	6.964
	15.811	1.581	0.770	0.982	3.044	0.976	0.762	0.972	8.015	6.892
		7.906	0.770	0.982	3.517	1.058	0.763	0.970	8.133	6.857
		15.811	0.773	0.982	4.044	1.156	0.772	0.984	8.526	7.006
		31.623	0.770	0.982	4.934	1.327	0.767	0.979	8.958	6.976
	31.623	1.581	0.640	0.965	3.873	1.176	0.643	0.973	8.284	6.963
		7.906	0.640	0.965	4.284	1.248	0.636	0.953	8.583	7.016
		15.811	0.636	0.964	4.729	1.332	0.637	0.967	8.736	6.930
		31.623	0.642	0.965	5.534	1.486	0.647	0.967	9.230	6.967
2	1.581	1.581	1.934	1.996	2.139	0.832	1.925	1.986	7.850	6.907
		7.906	1.935	1.996	2.781	0.929	1.934	1.992	8.032	6.952
		15.811	1.935	1.996	3.417	1.035	1.933	1.996	8.352	6.967
		31.623	1.935	1.996	4.432	1.224	1.926	1.984	8.805	6.930
	7.906	1.581	1.733	1.982	4.203	1.170	1.739	1.996	8.559	6.951
		7.906	1.728	1.981	4.569	1.244	1.733	1.985	8.839	7.045
		15.811	1.725	1.982	4.992	1.327	1.716	1.974	9.126	7.030
		31.623	1.723	1.982	5.757	1.480	1.728	1.983	9.568	7.072
	15.811	1.581	1.540	1.964	5.736	1.498	1.538	1.960	9.395	7.024
		7.906	1.540	1.964	6.017	1.557	1.549	1.974	9.601	7.040
		15.811	1.537	1.964	6.328	1.624	1.526	1.949	9.813	7.094
		31.623	1.540	1.964	6.942	1.748	1.537	1.955	10.202	7.075
	31.623	1.581	1.281	1.929	7.543	2.003	1.281	1.932	10.544	7.104
		7.906	1.289	1.931	7.793	2.050	1.284	1.929	10.830	7.277
		15.811	1.286	1.930	8.037	2.095	1.288	1.936	10.935	7.194
		31.623	1.293	1.931	8.472	2.191	1.292	1.930	11.316	7.279
4	1.581	1.581	3.869	3.992	3.794	1.097	3.858	3.986	8.560	7.044
		7.906	3.868	3.992	4.188	1.171	3.866	3.990	8.710	6.947
		15.811	3.870	3.993	4.629	1.258	3.864	3.991	8.927	6.999
		31.623	3.869	3.993	5.414	1.420	3.872	3.993	9.473	7.085
	7.906	1.581	3.461	3.964	8.188	1.986	3.457	3.962	11.179	7.205
		7.906	3.452	3.964	8.363	2.028	3.448	3.963	11.329	7.220
		15.811	3.445	3.963	8.668	2.084	3.449	3.963	11.514	7.245
		31.623	3.453	3.963	9.065	2.183	3.460	3.974	11.851	7.233
	15.811	1.581	3.079	3.929	11.310	2.714	3.082	3.934	13.636	7.513
		7.906	3.079	3.928	11.433	2.756	3.080	3.929	13.809	7.553
		15.811	3.088	3.929	11.585	2.794	3.090	3.934	13.831	7.420
		31.623	3.074	3.928	11.915	2.865	3.064	3.915	14.078	7.450
	31.623	1.581	2.558	3.858	14.906	3.801	2.562	3.861	16.696	7.990
		7.906	2.566	3.860	15.075	3.824	2.563	3.852	16.789	7.978
		15.811	2.563	3.860	15.271	3.855	2.558	3.854	16.975	7.981
		31.623	2.565	3.859	15.416	3.896	2.556	3.856	17.098	7.976

6 Results of Pre-pairing with Wavelets

Table 8: Returns Comparison of Pairing Methods when the Wavelet Transform is Applied Before the Pairing

Cointegration						
	Wavelet based Johansen			Standard Johansen		
period	Sym22	Standard	# of pairs	Sym22	Standard	# of pairs
1	21.28%	3.72%	5367	13.53%	-0.04%	4739
2	9.47%	-3.97%	5941	10.09%	-3.42%	5258
3	13.11%	-4.82%	4277	10.70%	-6.50%	2720
4	13.14%	-0.49%	5793	4.80%	1.43%	3614
5	9.78%	2.47%	6556	8.21%	1.61%	5692
6	10.43%	-0.17%	4437	8.78%	-3.62%	4817
7	8.63%	-4.97%	4208	11.50%	-2.12%	3600
average	12.26%	-1.18%	5225.571	9.66%	-1.81%	4348.571
Minimum Distance						
	Wavelet Based Min. Dist.			Standard Min. Dist.		
period	Sym22	Standard	# of pairs	Sym22	Standard	# of pairs
1	18.32%	2.25%	1000	13.58%	1.24%	1000
2	10.63%	-0.37%	1000	10.93%	-1.80%	1000
3	13.32%	-1.86%	1000	14.69%	-1.07%	1000
4	9.95%	1.63%	1000	7.97%	-0.33%	1000
5	10.78%	-0.49%	1000	13.63%	0.80%	1000
6	8.83%	-1.65%	1000	10.62%	0.69%	1000
7	9.82%	-4.28%	1000	11.29%	-3.42%	1000
average	11.66%	-0.68%	1000	11.82%	-0.55%	1000

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